MSDS 682 · DATA STREAM PROCESSING

Space Weather Radio-Risk Monitor

A reproducible Kafka alert when planetary Kp crosses an elevated-risk threshold.

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A radio operator needs a signal, not another feed to watch

$Kp \geq 6$

Our course-scale threshold for elevated radio-risk conditions.

Emit one alert when the rolling 15-minute maximum first reaches or exceeds 6.

- Target user: amateur-radio operators relying on HF propagation.
- No repeated alerts while the window remains elevated.
- The rule rearms only after the rolling maximum falls below 6.

DATA AND EVENT CONTRACT

Every NOAA observation becomes one ordered Kafka event

Source: NOAA SWPC one-minute estimated planetary K-index JSON.

Topic: `kp_observations`
Key: `planetary_kp`

Validation: UTC timestamps, Kp from 0-9, approved provenance, exact four-field contract.

The alert demo uses a clearly labeled synthetic replay because live conditions were quiet. A committed raw NOAA sample verifies the source shape.


```json
{
  "time_tag": "2026-08-11T00:10:00Z",
  "kp_value": 6.3,
  "source": "synthetic://kp-threshold-fixture",
  "ingested_at": "2026-08-11T00:10:05Z"
}
```


Representative record: the first event that creates an alert.

IMPLEMENTED ARCHITECTURE

Kafka separates validated ingestion from stateful alert logic

Replay
9 labeled JSONL events
including 1 duplicate

→

Producer
Validate first
publish in event order

→

Kafka
1 topic · 1 partition
constant global key

→

Consumer
Deduplicate
15-minute rolling max
crossing + rearm

→

Evidence
alert.json · metrics.csv
briefing.txt · evaluation.json

Why this architecture: Kp is one global ordered series, so one partition and one constant key preserve sequence. Kafka keeps the producer and consumer independent and makes replay and duplicates observable.

LIVE DEMONSTRATION

Leave the slides here

Switch to the local terminal, then show the UI.

1 · Terminal

Run `./run_demo.sh`
Point out real Kafka startup and replay.

2 · Result

Show 9 messages, 1 duplicate, and exactly 2 alerts.

3 · UI

Open the local UI and show the alert, metrics, and natural-language briefing.

The alert path is deterministic; AI only explains the result

9

Kafka messages processed, including one deliberate duplicate.

2

Expected threshold-crossing alerts, with no alert spam.

7/7

Evaluation assertions passed for alerts and AI briefing cases.

Deterministic code decides

Rolling maximum · threshold · label · alert · fallback.

→

Optional AI phrases five facts

Two sentences only. Changed numbers, labels, causes, places, or advice are rejected.

Verified: 167 standard tests; all 3 Docker-backed Kafka integration tests also passed separately.

Deterministic replay made the streaming behavior provable

WHAT WE LEARNED

Quiet live data could not demonstrate a threshold crossing. A labeled replay reliably proves crossing, deduplication, window expiry, and rearming.

LIMITATION

This is a recent global Kp condition - not a forecast or station-level propagation model.

REALISTIC NEXT STEP

Add a long-running consumer that continuously processes live NOAA events and refreshes the UI.

The result is small, explainable, and reproducible: one ordered stream, one clear rule, and evidence for every alert.

BACKUP · USE ONLY IF THE LIVE DEMO FAILS

Saved terminal evidence

```
$ ./run_demo.sh
Published 9 event(s) · Topic=kp_observations · key=planetary_kp
Processed 9 valid event(s); emitted 2 alert(s)
Briefing source: fallback
Evaluation: 7/7 assertions passed
Artifacts match the labeled fixture:
  2 alerts · 9 metrics rows · briefing present
167 passed, 3 opt-in tests skipped
Demo complete.

Artifacts
  outputs/alert.json
  outputs/metrics.csv
  outputs/briefing.txt
  evaluation/evaluation.json
```